CarMilan

PROJECT DOCUMENTATION

Requirements • Features • Workflows • Architecture • Database • Security • Testing

“Buy Together. Negotiate Better.”

Version 1.0 | 26 October 2026
Build guide for Antigravity

Document Control

• Product requirements and implementation specification

• CarMilan — group car-buying and dealer quotation platform

• Initial market: India

• Audience: project owner, developers, QA testers, AI coding agents

• Initial baseline; see open decisions before launch

How to use this document

Treat this as the project source of truth. Implement in phases, validate each requirement with tests, and document changes. All sample prices and dealer records are illustrative unless independently verified.

1. Executive Summary

CarMilan is a group car-buying platform for the Indian market. Buyers interested in the same car model/variant can form a group, request quotations from eligible dealerships, compare itemized offers, and independently decide what to do next. The platform organizes buyer demand; it does not guarantee discounts, vehicle availability, or final purchase prices.

Tagline: “Buy Together. Negotiate Better.” Working name: CarMilan. Check trademark, domain, and social-handle availability before launch.

2. Problem, Goals, and Non-Goals

Problem

Buyers often contact dealers one by one and compare inconsistent quotations. Dealers may struggle to find groups of buyers with similar purchase intent. CarMilan standardizes group interest and quotation comparison.

Goals

• Let buyers find cars and join/create groups.

• Let eligible groups request competitive dealer quotations.

• Compare offers using consistent price components and clear terms.

• Give verified dealerships a structured channel to respond.

• Protect personal information and avoid misleading claims.

Not in the MVP

• Vehicle payments, deposits, financing, or direct vehicle sales.

• Guaranteed discounts or guaranteed delivery.

• Unverified inventory integrations.

• Public exposure of buyers’ phone numbers or email addresses.

3. Scope and Priorities

MVP — Must Have

• Landing page and car catalog.

• Buyer authentication and profile.

• Create, browse, join, and leave groups under defined rules.

• Group capacity, deadlines, and status lifecycle.

• Dealer registration and admin verification.

• Quotation requests and dealer offer submission.

• Offer comparison with itemized prices and expiry.

• Buyer, dealer, and admin dashboards.

• Role-based access, database security, error states, responsive UI, and test coverage.

• Clearly labelled demo data.

Later Releases

• Email/SMS/WhatsApp notifications with consent and provider setup.

• Saved cars/groups, analytics, exports, multilingual UI, dealer appointment scheduling.

• Verified inventory or licensed reference-price integrations only when available.

4. User Roles and Permissions

Visitor

• Browse public pages and catalog; register or sign in. No private dashboard actions.

Buyer

• Manage own profile; create/join eligible groups; view group progress; request/track quotations where permitted; compare offers; save cars; receive notifications; report content.

• Cannot edit dealer offers or see other buyers' private contact details.

Dealer

• Submit business details for verification; manage own dealership; view eligible requests; submit/edit/withdraw own offers within rules; view own history.

• Cannot approve itself, edit another dealer's offers, or access buyer private details without a consent-based need.

Administrator

• Verify/suspend dealers, manage catalog and groups, moderate reports, review audit logs, and manage platform settings. Sensitive actions must be logged.

Authorization must be enforced on the server/database, not only by hiding UI controls.

5. Functional Requirements

FR-01 Authentication and profiles

• Register, sign in/out, recover account, persist sessions securely.

• Validate required fields; users may edit their own profile but cannot grant themselves admin privileges.

FR-02 Car catalog

• Store brands, models, variants, fuel, transmission, body type, model year, and sourced reference price if available.

• Search/filter by model, budget, fuel, transmission; distinguish indicative from verified live pricing.

FR-03 Create group

• Buyer selects variant, city/state, target size, purchase timeline, and optional notes.

• Validate required fields and configurable minimum/maximum group size.

• Record creator and timestamps; initial status is Recruiting.

FR-04 Join/leave group

• Only authenticated buyers can join Recruiting groups with capacity.

• Prevent duplicate membership with a database unique constraint.

• Enforce capacity safely during simultaneous join requests.

• Define creator-leaving rules; reject joins to full, locked, cancelled, or expired groups.

FR-05 Group lifecycle

• States: Recruiting, Target Reached, Quotations Requested, Offers Received, Completed, Cancelled, Expired.

• Validate every state transition server-side.

• Only set Offers Received when a valid offer exists; expire groups/requests reliably.

FR-06 Quotation request

• Eligible group creates request with variant, city, deadline, delivery window, and price components.

• Send to eligible active dealers matching brand/region rules.

• Prevent duplicate active requests unless opening a new round.

FR-07 Dealer verification

• Dealer submits business details and documents if required.

• Pending/rejected dealers cannot appear as verified or submit eligible offers.

• Admin approves/rejects with reason; restrict document access and record decision history.

FR-08 Dealer offers

• Verified dealer submits offer for matching request/variant.

• Include itemized costs, discount, estimated on-road price, validity, delivery estimate, and terms.

• Validate numbers, expiry, ownership, and allowed status transitions; preserve offer history.

FR-09 Offer comparison

• Compare matching variants; show line items, mandatory/optional costs, validity, delivery, and terms.

• Sort by quoted total, delivery estimate, discount, or expiry.

• Warn if offers are incomplete or not comparable.

• Show savings only against a valid, named reference price and disclosed formula.

FR-10 Notifications

• In-app notifications for milestones, new offers, expiry, and admin decisions; recipients can mark read.

• External email/SMS requires configured provider, failure handling, and relevant consent.

FR-11 Admin and moderation

• Filter/manage users, dealers, groups, requests, offers, and reports.

• Audit sensitive actions with actor, action, target, timestamp, and reason.

FR-12 Support and policies

• Provide support route, privacy notice, terms, and clear offer disclaimers.

• Do not claim partnerships unless verified and authorized.

6. Business Rules

• Joining a group is not a purchase contract or payment commitment.

• No buyer payment is needed for the MVP.

• Group size and eligibility are configurable, not hardcoded.

• Only active verified dealers can submit eligible offers.

• Offers are dealer-authored, timestamped, and time-limited.

• Optional accessories must be separated from mandatory charges.

• Avoid double-counting taxes, registration, insurance, discounts, or totals.

• Expired offers are visibly marked and excluded from current-offer sorting by default.

• Buyer contact information is private by default.

• Demo prices, dealers, testimonials, and counts must be clearly labelled as fictional.

• Do not guarantee discounts, inventory, delivery, or final transaction outcomes.

7. Core Workflows

Buyer joins a group

1. Browse car and select variant.

2. Sign up/sign in.

3. Filter groups by city, variant, status, and timeline.

4. Open a Recruiting group and review rules, capacity, and deadline.

5. Click Join Group.

6. Server checks authentication, status, capacity, duplicate membership, and eligibility atomically.

7. Save membership, calculate count from database records, and show confirmation.

Buyer creates a group

1. Choose variant, city, target size, timeline.

2. Validate all inputs and configured limits.

3. Create group and creator membership transactionally.

4. Display group in browse results with Recruiting status.

Quotation round

1. Eligible group opens a quotation request.

2. System records request and notifies eligible verified dealers.

3. Each dealer submits itemized offer and expiry.

4. Server validates and stores offer under that dealer.

5. Group/request status updates when a valid offer exists.

6. Buyers compare offers; no offer is automatically selected.

Dealer verification

1. Dealer submits profile and verification details.

2. Status remains Pending.

3. Admin reviews and approves/rejects with reason.

4. Only approved dealers can submit eligible offers.

8. Pages and Navigation

Public

• Landing page; How It Works; Browse Cars; Car Details; Explore Groups; Group Details with public-safe data; Dealer Information; About; Contact; FAQ; Login/Register/Recovery; Privacy; Terms.

Buyer

• Dashboard; Create Group; My Groups; Group Details; Quotation Status; Compare Offers; Saved Cars; Notifications; Profile.

Dealer

• Dashboard; Dealer Profile; Verification Status; Incoming Requests; Request Details; Submit/Edit Offer; My Offers; Offer History.

Admin

• Overview; Buyer Management; Dealer Verification; Dealer Management; Catalog Management; Group Management; Requests/Offers; Reports; Audit Logs; Settings.

9. UI/UX Requirements

• Use a clean, premium automotive/fintech design with readable typography, clear hierarchy, consistent spacing, and high-quality imagery used lawfully.

• Mobile-first responsive layouts; test common phone, tablet, and desktop widths.

• Reusable buttons, forms, cards, tables, status labels, dialogs, and notifications.

• Provide loading, empty, error, success, and disabled states.

• Use status text as well as color; ensure keyboard navigation, visible focus, labels, and contrast.

• Confirm destructive actions. Format currency in INR and support Indian city/state fields.

• No fake testimonials, fake live counts, fabricated discounts, or unverified dealer logos.

10. Technical Architecture

Recommended stack: React + TypeScript + Vite; Tailwind CSS; Supabase Auth and PostgreSQL; database Row Level Security (RLS); server-side functions for sensitive operations; Lucide icons; optional Recharts for analytics.

Architecture rules

• Inspect the existing repository before replacing code.

• Use typed models, centralized validation, database constraints, and migrations.

• Use transactions/server functions for group joins, group creation, price calculations, and status changes.

• Never trust client-side role checks or price totals.

• Keep service-role keys and secrets out of frontend code.

• Separate development, staging, and production databases/data.

• Use a .env.example with variable names only.

Suggested structure

```
src/app/ # routes, providers, access guards\nsrc/components/ # reusable UI\nsrc/features/auth/\nsrc/features/cars/\nsrc/feat
ures/groups/\nsrc/features/quotations/\nsrc/features/dealer/\nsrc/features/admin/\nsrc/features/notifications/\nsrc/lib/ #
client, validation, helpers\nsrc/types/ # shared types\nsrc/tests/ #
tests\nsupabase/migrations/\nsupabase/functions/\nsupabase/seed.sql\ndocs/ # requirements, architecture, test plan
```

11. Database Requirements

Table	Purpose / key fields
profiles	auth user ID, display name, phone, city/state, contact preferences
user_roles	user ID, buyer/dealer/admin; server-controlled
dealerships	owner, business name, address, brands, verification status
dealership_verifications	document metadata, status, reviewer, reason, timestamp
car_brands	brand name, active
car_models	brand ID, model name, body type, active
car_variants	model ID, trim, fuel, transmission, year, optional sourced reference price
buyer_groups	creator, variant, city/state, target, status, timeline, deadline
group_memberships	group, buyer, status, joined/left timestamps; unique group+buyer
quotation_requests	group, creator, status, deadline, delivery window
dealer_offers	request, dealership, variant, status, validity, delivery, terms
offer_line_items	offer, component type, label, amount, optional flag, notes
saved_cars	user ID, variant ID; unique pair
notifications	recipient, type, title, message, read timestamp
audit_logs	actor, action, entity, reason, metadata, timestamp

Constraints and indexes

• One membership per buyer/group; enforce active membership rules.

• Target size within configured bounds; amounts validated.

• Offer variant must match request variant.

• Dealers can mutate only their own offers.

• Only authorized roles can change protected status fields.

• Index groups by status/city/variant, requests by status/deadline, offers by request/dealer/status, and notifications by recipient/read state.

• Keep verification files in private storage; do not use public URLs.

12. State Machines

Group state	Allowed next states	Condition
Recruiting	Target Reached, Cancelled, Expired	Target met, cancellation, or deadline
Target Reached	Quotations Requested, Cancelled, Expired	Quotation round opens or group closes
Quotations Requested	Offers Received, Completed, Cancelled, Expired	Valid offer or closure/expiry
Offers Received	Completed, Cancelled, Expired	Group concludes or closes
Completed	None	Terminal
Cancelled	None	Terminal
Expired	None	Terminal

Offer states: Draft → Submitted → Updated/Withdrawn/Expired. Optional “Selected” status must mean buyer interest only unless a separate legally reviewed transaction flow exists. Keep state transitions server-side and audit meaningful changes.

13. Price Calculation and Comparison

Store price components separately: ex-showroom, registration/road tax, insurance, mandatory fees, dealer discount, optional accessories, and other disclosed charges. Define whether each component is already included in another total.

• Use decimal-safe currency calculations; avoid floating-point money errors.

• Do not add registration/tax again if included in a supplied on-road figure.

• Keep optional accessories outside mandatory total unless selected.

• State whether insurance is dealer-provided, buyer-selected, or estimated.

• Show component source, quote date, and expiry.

• Warn when offers have different assumptions.

• Savings = valid documented reference price minus comparable quoted price; otherwise do not show savings.

14. Security, Privacy, and Compliance

• Use a trusted auth provider and secure session handling.

• Apply least privilege and RLS to every user-facing table.

• Validate on the server; protect against injection, XSS, insecure uploads, and unauthorized object access.

• Rate-limit sensitive operations and log important security events.

• Keep secrets server-side and rotate exposed credentials.

• Collect only necessary personal data and explain use in a privacy notice.

• Do not expose buyer contact details by default.

• Restrict verification documents and audit sensitive access.

• Define retention/deletion and support procedures.

• Get consent for marketing/third-party messaging as applicable.

• Review relevant Indian privacy, consumer protection, advertising, and e-commerce obligations with qualified counsel before launch.

• Clearly state that CarMilan facilitates comparison; dealers remain responsible for their quotations and final sale terms.

15. Non-Functional Requirements

Category	Requirement
Usability	Buyer can understand how to find a car and join a group without assistance.
Responsive	Core workflows work on mobile, tablet, and desktop.
Performance	Set measurable budgets; paginate lists and measure real load times.
Reliability	Handle retries, duplicate submissions, network failure, and partial notification failures safely.
Security	Database-enforced authorization, private data access, secure secrets, audit logs.
Accessibility	Semantic HTML, keyboard navigation, visible focus, contrast, labels, accessible errors.
Maintainability	Reusable components, typed APIs, linting, migrations, documented setup.
Scalability	Indexed queries and paginated group/offer lists.
Observability	Useful error logs without leaking private data.
Compatibility	Test mainstream desktop and mobile browsers.

16. Testing and Acceptance Criteria

Acceptance criteria

• Buyer can register/login and update own profile.

• Buyer can create a valid group and see it in listings.

• Duplicate memberships are blocked.

• Concurrent joins never exceed group capacity.

• Expired/cancelled groups cannot be joined.

• Pending dealers cannot submit eligible offers.

• Verified dealer can submit a valid offer for matching variant.

• Dealer cannot edit another dealer’s offer.

• Totals match documented price formula.

• Expired offers are clearly marked.

• Unauthorized users cannot access admin actions or private documents.

• All buttons/forms work and show useful error states.

• Demo records are never presented as real offers.

Test case examples

ID	Scenario	Expected result
TC-01	Valid registration	Account created; sign-in works.
TC-02	Invalid/missing form fields	Validation shown; no invalid record.
TC-03	Group target below minimum	Request rejected.
TC-04	Join group with capacity	One membership created.
TC-05	Join same group twice	Duplicate prevented.
TC-06	Two users take final slot	Capacity not exceeded.
TC-07	Join expired group	Join rejected.
TC-08	Pending dealer submits offer	Blocked.
TC-09	Offer uses wrong variant	Rejected.
TC-10	Dealer edits competitor offer	Access denied.
TC-11	Invalid negative price	Blocked unless documented adjustment type.
TC-12	Offer expires	Marked expired.
TC-13	Buyer opens admin route	Access denied.
TC-14	Read another buyer’s private fields	Private data not returned.
TC-15	Network error during join	Clear retry state; no duplicate.
TC-16	Incomplete offer comparison	Warning displayed.
TC-17	Mobile viewport	No clipped controls; workflow usable.
TC-18	Keyboard-only navigation	Controls reachable; focus visible.

Definition of Done

• Build, lint, type checks, and relevant tests pass.

• Migrations run on a clean database.

• Role/RLS policies tested with separate accounts.

• Critical/high defects resolved or explicitly documented.

• Actual test results recorded; no invented pass results.

• README, environment, and deployment instructions complete.

17. Edge Cases

• Two buyers attempt the final slot at the same time.

• Group creator leaves after others join.

• Group deadline passes during a quotation round.

• Dealer verification is revoked while offers are active.

• Dealer updates an offer already viewed by buyers.

• No offers arrive or all offers expire.

• Duplicate/rapid submissions occur.

• Connection drops during a mutation.

• Database save succeeds but notification delivery fails.

• Variant is deactivated while a group references it.

• Buyer changes city/timeline after joining.

• User is suspended while memberships/offers exist.

Define a policy for each case, record important state changes, and show a clear message. Never silently discard data.

18. Demo Data Policy

• Use fictional sample buyers/dealers for demos.

• Label seeded prices/offers “Demo data — not a real quotation.”

• Do not use real logos or imply partnerships without permission.

• Do not fabricate savings, testimonials, live inventory, or participant counts.

• Provide seed/reset scripts for development and keep demo data out of production.

19. Deployment Checklist

• Separate development, staging, and production environments.

• Configure Supabase project and auth redirect URLs.

• Apply migrations and test RLS policies.

• Set environment variables in hosting; keep secrets out of Git.

• Run seed data only in development/demo.

• Configure backups and test restoration.

• Set up logs/error reporting as appropriate.

• Use HTTPS; check domain availability before selecting a domain.

• Test buyer, dealer, and admin workflows in staging.

• Review legal pages and obligations before public release.

• Document rollback steps for failed deployments/migrations.

20. Build Roadmap

Phase	Deliverables	Exit condition
1. Foundation	Repository, routing, design system, responsive shell	App builds; navigation/layout work.
2. Catalog	Brands, models, variants, search/filter	Browsing uses persistent data.
3. Identity	Auth, profiles, roles, RLS	Role boundaries tested.
4. Groups	Create/join/leave, capacity, lifecycle	Group and concurrency tests pass.
5. Dealer onboarding	Profile and admin verification	Only approved dealers submit offers.
6. Quotations	Requests, line items, validation	Offers persist and match requests.
7. Comparison	Offer table, price math, expiry	Totals correct; assumptions visible.
8. Admin/notifications	Notifications, moderation, audit	Sensitive operations logged and authorized.
9. QA/release	Tests, fixes, deployment docs	Critical acceptance criteria pass in staging.

Build and verify one phase at a time. Avoid asking an AI coding agent to build every feature in one unverified pass.

21. Antigravity Instructions

First prompt

```
Read docs/CarMilan_Project_Documentation.md completely. Inspect the existing repository and report the current stack, implemented features, and gaps. Implement Phase 1 only. Preserve working code. Run available build, lint, type-check, and tests; report actual results and unresolved issues.
```

Prompt for each next phase

```
Implement the next phase in the CarMilan documentation. Preserve working features. Add or update migrations, authorization policies, validation, tests, and docs as needed. Execute relevant tests and fix failures. Do not fabricate test results, real dealer data, or live integrations. Summarize changed files, test results, setup steps, and limitations.
```

22. Glossary

Term	Meaning
Buyer group	Users interested in the same car configuration and region.
Variant	Specific trim/configuration of a car model.
Quotation request	Structured request asking eligible dealers for offers.
Dealer offer	Dealer-authored, time-limited quotation with itemized terms.
On-road price	Estimate including applicable costs; exact components must be explained.
RLS	Row Level Security: database policies controlling access to rows.
MVP	Minimum Viable Product: smallest useful version testing the core concept.
Audit log	Record of important actions, who performed them, and when.

23. Decisions Before Public Launch

• Which cities and car brands launch first?

• Who can create groups and open quotation rounds?

• What minimum group size makes a request eligible?

- How are dealers verified and verification renewed?
- Which price components are mandatory?
- When do group members see offers?
- What is the group cancellation policy?
- Which notification channels and consents are required?
- How are inaccurate quotations disputed?
- What data retention/deletion rules apply?
- Is there a licensed source for reference prices?
- What legal review is needed for the business model?

Recommended defaults: launch in a small number of cities, use a manually curated catalog, verify dealers manually, avoid payments, show eligible submitted offers transparently to group members, and label every estimate/demo record.

24. Final Product Definition

The core CarMilan workflow is: choose a car → join a group → reach eligibility → request dealer quotations → compare itemized offers → make an independent decision. Prioritize trust, accurate pricing, secure role permissions, and a reliable end-to-end workflow over decorative features.